

Andrea Gemelli

AI Research Scientist & Engineer

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andreagemelli.me | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [Hugging Face](#)

Professional Summary

AI Research Scientist with a **PhD in Document AI and Natural Language Processing (NLP)** and 6+ years bridging research and production across **Machine Learning** and **Deep Learning**. I publish at top-tier venues (**ECCV**, **ICASSP**, **IJDAR**) and ship the results: training and deploying **Vision and Language Models (VLMs)**, building end-to-end ML pipelines, and turning state-of-the-art (SOTA) **LLM** and agentic-systems research into reliable production components. Expert in **Python** and **PyTorch**. Research has reached **150+ citations**; open-source models and datasets have surpassed **20K+ downloads** on Hugging Face. Third place at the **{Tech: Europe} AI Hackathon** in Paris among 120 applicants.

Experience

AI Research Scientist — [Kleio](#)

June 2026 – Present

Paris, France

- Design and evaluate **multi-agent architectures** for an intelligent agent-orchestration platform.
- Advance **retrieval-augmented (RAG) and orchestration pipelines**, translating recent LLM and agentic-systems research into production-ready components.
- Partner with product and engineering teams to define the research roadmap and establish rigorous applied-research practices at team scale.

AI Research Scientist — [LetXbe](#)

Dec 2023 – April 2026

Paris, France

- Trained a proprietary **document information-extraction VLM** matching commercial-API performance while preserving data privacy.
- Fine-tuned (**QLoRA**) and benchmarked open-source VLMs (**up to 32B parameters**) on proprietary DocVQA datasets of **up to 100K documents**.
- Built an **end-to-end ML pipeline** automating preprocessing, data-quality checks, training, evaluation, and deployment; raised production precision from **85% to 95%**.
- Drove **3× business-pipeline growth** through VLMs in production.
- Led the R&D roadmap, supervised 3 interns (resulting in published papers), and ran a monthly internal reading group.
- Open-sourced fine-tuned Qwen and Mistral models for DocVQA and the **BoundingDocs** dataset (20K+ downloads on Hugging Face).

Education

PhD in Document AI — University of Florence

2020 – 2023

Florence, Italy & Barcelona, Spain

- Awarded the **Doctor Europaeus** title (Feb 2024) for a thesis on Document AI and Graph Neural Networks.
- Published **8 peer-reviewed papers** in top-tier venues and journals (ECCV; oral at ICASSP; IJDAR and PRL).
- **Visiting Researcher** at the [Computer Vision Center](#) (Barcelona) for one year, leading to a publication at ECCV.
- Summer schools: **Cambridge ELLIS 2022** (50% acceptance) and **ICVSS 2023** (30% acceptance).

MSc in Computer Science and Engineering — University of Florence

2017 – 2020

Florence, Italy

- Graduated with **full marks (110/110)**. Thesis completed during an Erasmus+ Traineeship at the L3i lab, University of La Rochelle.
- Erasmus+ Study exchange at Ghent University; coursework in Machine Learning, Software Hacking, and Parallel Computing.
- Data Scientist at [Alias2k](#), developing AI solutions to enhance company products.

Teaching

- **Lecturer** — **AI for Design**, University of Florence (2024): design and deliver master-level coursework on AI/ML tools and applications, including lectures on LLMs, agents, and Document AI.
- Teaching assistant for university programming courses during the PhD; delivered AI lectures and supervised BSc/MSc theses.

Publications

Full list on [Google Scholar](#) (150+ citations).

- **IJDAR 2025: BoundingDocs: a Unified Dataset for Document Question Answering with Spatial Annotations.**
- **ICCV 2025 (VisionDocs Workshop, Poster):** Towards Reliable and Interpretable Document Question Answering via VLMs (arXiv).
- **ICASSP 2024 (Oral):** Structure Matters: Analyzing Videos Via Graph Neural Networks for Social Media Platform Attribution.
- **IJDAR 2024:** Datasets and annotations for layout analysis of scientific articles.
- **PRL 2023:** Automatic generation of scientific papers for data augmentation in document layout analysis.
- **DocEng 2023:** Deep-learning for dysgraphia detection in children handwritings.
- **IRCDL 2023:** CTE: A Dataset for Contextualized Table Extraction.
- **ECCV 2022 (TiE Workshop):** Doc2Graph: a Task-Agnostic Document Understanding Framework Based on Graph Neural Networks.
- **ICPR 2022:** Graph Neural Networks and Representation Embedding for Table Extraction in PDF Documents.
- **S+SSPR 2022:** Data Augmentation on Graphs for Table Type Classification.

Achievements & Community

- Active reviewer for the **IJDAR** journal and **ICDAR** conference.
- **Third place** at the {Tech: Europe} AI Hackathon (Paris) with **EasyMCP**, translating OpenAPI specs to MCP servers.
- Built **Look 4 Fundings** (an MCP server + app for EU funding discovery, integrated with Le Chat) at the Mistral AI MCP Hackathon, where it was highlighted by **Sophia Yang** (Mistral AI).
- **Third place** at the Wacom x Luleå University hackathon with a handwriting-labeling game.
- **IAPR Research Scholarship (IAPR RS)**, awarded by the IAPR through its Education Committee (IAPR-EC), 2023.
- **200+ GitHub stars** across open-source projects, including **Doc2Graph**, a framework for Document AI.
- Author of an AI/tech blog on research and ML best practices, published on my [portfolio](#).

Skills & Tools

ML & Modeling: PyTorch, Transformers, TRL, PEFT, vLLM, Ollama, Gradio, NumPy, Pandas, Scikit-Learn

Dev & Tooling: Python (Expert), TypeScript, Git, Docker, FastAPI, FastMCP, UV (ruff & ty), coding harnesses (Claude Code, OpenCode)

MLOps & Experimentation: CometML, Weights & Biases, training/eval monitoring, data-quality pipelines

Cloud & Infrastructure: Google Cloud (Vertex AI, VMs, Buckets), AWS (SageMaker, S3, Lambda), Scaleway

Languages: Italian (Native), English (C1), French (B1), Spanish (B1)